# Field Determination of Resilient Pressure Stiffness (k_p)

Condensed operating procedure and geometric reference sheet

This briefing turns the original note into a print-first field handout. Key equations are isolated into bordered math cards and the procedure is condensed into an operational checklist.

- Plate Diameter: 0.3 to 0.45 m
- Target Pressure: 1g tire bearing pressure
- Typical Cycling Rate: 0.5 to 1.0 Hz

FIELD PROCEDURE

## Execution sequence

**01 Prepare and place the plate**

Select representative haul-road locations, smooth the immediate contact zone, and seat the bearing plate squarely on the running surface.

**02 Match the target tire pressure**

Set the cyclic loading target to the nominal 1g tire bearing pressure for the haul truck being modeled.

**03 Cycle until deformation converges**

Repeat unload-reload cycles until the resilient portion of the deformation stabilizes and the response becomes repeatable.

**04 Record the resilient rebound response**

Use the peak cyclic stress and the resilient elastic displacement from the converged cycle to compute k_p.

EQUIPMENT

## Required field setup

- Cyclic plate load assembly with known footprint area.
- Hydraulic ram with reliable pressure readout.
- Independent ground-referenced displacement measurement.
- Data acquisition system to confirm cyclic convergence.

PRIMARY STIFFNESS EQUATION

### Resilient pressure stiffness

$$k_p = \frac{\Delta\sigma}{\Delta\delta_{\text{resilient}}}$$

Use the peak cyclic pressure and the resilient elastic deformation measured from the converged cycle.

FOOTPRINT AREA

### Empirical tire contact relation

$$A = 1.35 \times \delta_{tire} \times \phi$$

The area is inferred from tire deflection and diameter rather than measured directly in the field.

FOOTPRINT CHORD

### Contact patch length

$$L = 2\sqrt{(\frac{\phi}{2})^2 - (\frac{\phi}{2} - \delta_{ti}}$$

This is the cleanest print form of the chord equation because it exposes the underlying circle geometry directly.

INTERPRETATION NOTES

### How the geometry feeds the rolling resistance model

- The simulator only needs a few primary drivers: tire load, tire stiffness, tire diameter, and resilient ground stiffness.
- From those values it can derive tire deflection, footprint area, footprint chord length, and the geometric interaction arc.
- Separating the equations into isolated math cards avoids the dense inline notation that becomes difficult to read in PDFs.

Prepared as a compact field reference for the haul-road rolling resistance workflow.